MagicBox: Curriculum Plan

How the 27 activities fit together — sequencing, timing and equipment

Level: Ages 7–19 • **Time in class:** From a single lesson to a full school year • **Sensors:** all four: light, temperature-humidity-pressure, CO_2, motion

1. What this collection is

MagicBox is a sensor platform for school science. A small box containing four sensors streams live measurements to a laptop application. The application's **Activities** page sorts them into three groups:

- **15 guided experiments** (in the guides also *guided exercises*) — ordered steps that walk a class through a specific investigation, with theory, predictions, measurements, a "How sure are we?" step and a journal to record observations. These are lesson-shaped: start, middle, end.
- **8 explore tools** — dashboards for open investigation and long-running projects.
- **4 games**, where the box is a control stick.

Every activity has **two documents**: a **teacher guide**, covering curriculum links, a time plan, the materials list, the science background, the lesson flow, analysis and discussion questions, common pitfalls, safety notes, adaptations and resources; and a separate **pupil sheet**, two pages long, printed double-sided, one per child. Both exist **in Romanian, English, French and German**; the Romanian version is the one written first. The same goes for the application: it starts in English, and Romanian (or French or German) is chosen in **Settings**.

The one thing that matters most

Whatever activity you choose, ask pupils to **write down a prediction before any measurement begins**, with a reason. The pedagogical value of these boxes lies almost entirely in the gap between what pupils expect and what the sensor reports. A measurement that merely confirms an unstated expectation teaches very little; a measurement that contradicts a written prediction is remembered for years. The guided experiments help here: before each measurement the application asks for the written prediction, and at the conclusion it quotes it back to the pupils.

2. The three levels

Activities are labelled by level inside the app. The levels describe *cognitive demand* rather than strict ages, and a capable younger class can attempt the next level up with support.

Level	Ages	What is expected of pupils
Junior	7–10	Read a number, compare two numbers, describe a change in words. Qualitative reasoning: bigger, smaller, faster, warmer.
Explorer	11–14	Read a graph, identify a trend, control a variable, take a systematic series of readings, express a simple relationship.
Scientist	15–19	Calculate a rate or gradient, transform data (logarithms, from grade X), fit a line, evaluate assumptions and sources of error, connect to formal theory.

Two mathematical thresholds decide, more than age, how far a class can go: **the cosine**, which

arrives in grade VII, and **logarithms**, which arrive in grade X. All Scientist-level content that uses log or ln is marked "for grade X and up" in the guides; in grade IX the same activities are done with ratios, percentages or the linear rule.

3. The full catalogue

3.1 Guided experiments

The *Time* column is the measurement phase only, as on the card in the application; the *Grades* column says who the activity is designed for; the *Steps* column counts every step in the application, the predictions and the "How sure are we?" step included.

Activity	Level	Grades	Time	Steps	Core idea
Light is Everywhere	Junior	II–IV	15 min	6	Light varies enormously between places; measurement beats intuition.
The Magic Thermometer	Junior	II–IV	20 min	6	Temperature differs from place to place and changes over time.
The Air We Breathe	Junior	II–IV	15 min	7	Breathing changes the air; CO_2 is invisible but measurable.
Flying Straight	Junior	II–IV	15 min	6	Orientation and tilt can be measured.
The Box in Balance	Junior	II–IV	15 min	7	Gravity gives a reference direction.
Indoor Pollution	Explorer	V–VIII	20 min	8	CO_2 indicates ventilation quality.
Air Before and After Airing	Explorer	V–XII	45 min	8	CO_2, particles and vapours do not tell the same story about the air (optional sensors).
Soil and Water	Explorer	V–XII	50 min	7	Soils hold water differently; a probe is calibrated at two points (optional soil probe).
The Seismometer	Explorer	V–XII	20 min	7	Vibrations can be recorded and read.
Pressure and Altitude	Explorer	V–VIII	20 min	7	Pressure falls by about 0.12 hPa for every metre climbed; the formula with an exponent stays an advanced note.
Urban Heat Islands	Explorer	V–VIII	25 min	11	Built surfaces are warmer than green ones.
Air Convection	Explorer	V–VIII	20 min	8	Warm air rises; gradients form.
The Water Cycle	Explorer	V–VIII	20 min	9	Evaporation and condensation, humidity; the application computes the dew point.
CO_2 and Photosynthesis	Scientist	X–XII	50 min	11	Net versus gross rates; inference from two measurements. Needs **at least 45 minutes of measurement**.
Solar Panels	Scientist	IX–XII	30 min	10	Incidence angle governs collected energy; the application has a step for the tilt-angle series.
Physics of the Atmosphere	Scientist	IX–XII	50 min	10	The barometric formula; the scale height, $H = -1/m$, from the gradient of the $\ln P$ against height line, plotted in a spreadsheet from the exported data (about 8 km; grade X and up).
The Beer–Lambert Law	Scientist	X–XII	50 min	9	Exponential attenuation; linearising by taking logarithms. Enrichment, not core curriculum.

3.2 Explore tools and games

Activity	Type	Grades	Core idea
Measurement Notebook	Instrument	IV–XII	Systematic labelled spot readings.
Weather Station	Instrument	V–XII	Live atmospheric monitoring.
Air Quality	Instrument	V–XII	Classroom CO_2, several boxes at once.
Solar Energy	Instrument	VI–XII	Light as an energy resource.
Seismology	Instrument	V–XII	A working seismograph.
Heat Islands with Several Boxes	Instrument	V–XII	Temperature in several places at once, one box in each.
The Greenhouse Effect	Instrument	V–XII	CO_2 and light together.
Spectroscopy	Instrument	IX–XII	Transmission and absorption of light.
Among the Clouds	Game	II–VIII	Tilt control; the accelerometer senses gravity.
Earthquake!	Game	IV–XII	Real earthquake safety; P waves precede S waves; the door-frame myth.
Resonance	Game	XI	Natural frequency; resonance comes from physics, not from force.
Gyro Target	Game	V–XII	Precision, steadiness, and how a reflex is relearned.

4. Activities by grade and subject

This section shows where each activity enters the Romanian school curriculum: first a few placements worth knowing, then the table by grade and subject. The same data, on a single page to pin up, is in `docs/en/activities_by_grade_and_subject.pdf`, with the file name of every guide. The *Grades* column in the catalogue says who the activity is designed for; the cells of the table say where it meets a curriculum lesson.

4.1 Placements worth knowing

- **Photosynthesis** is taught in **Biology in grades V–VI** (here, as a demonstration: CO_2 falls in the light) and in **grade X** (with rates and the compensation point). It is not a curriculum entry in grade IX, nor in grades XI–XII.
- **Earthquakes** are taught in **Geography in grade V** and in **grade IX**, not in grade VI. Quantitative seismic waves: Physics, in upper secondary.
- **Resonance** and oscillations are taught in **Physics in grade XI**; that is why the app lists the *Resonance* game at the Scientist level only (ages 16–18).
- **The Beer–Lambert law** and colorimetry are **enrichment** or olympiad preparation, not core curriculum. The logarithm they use belongs to Mathematics, grade X.
- **The cosine** arrives in **grade VII**, so the angle series in *Solar Panels* and *Solar Energy* can be compared with the cosine law from there up.
- **Logarithms** arrive in **grade X**. All Scientist-level content with log or ln — the scale height in *Physics of the Atmosphere*, absorbance in *Spectroscopy*, *The Beer–Lambert Law*, the logarithmic scales discussed in *The Seismometer* — is for grade X and up.
- **Pressure at Explorer level** is worked with the rule $\Delta P \approx 0.12$ **hPa per metre** (1.2 hPa per 10 m) and the difference between floors; the barometric formula with an exponent is an advanced note. Atmospheric pressure is taught in Physics in grade VII.
- **Heat transfer** (conduction, convection, radiation) is taught in **Physics in grade VIII**.

- **Evaporation and condensation** appear in Natural Sciences (grades III–IV) and in lower-secondary Physics, but **we are not sure** whether the programme in force places them in grade VI or grade VIII; the table therefore marks both with a question mark. Check the programme in force. Also marked with a question mark: inertia in *The Seismometer* (Physics, grade VII).

4.2 The table by grade and subject

Activity	II–IV	V	VI	VII	VIII	IX	X	XI–XII
Guided experiments								
Light is Everywhere	S M							
The Magic Thermometer	S M							
The Air We Breathe	S M							
Flying Straight	S M							
The Box in Balance	S M							
Indoor Pollution		G	M P	B	P			
Air Before and After Airing		G	M	B	P	G		
Soil and Water		G	B	B		G	B	
The Seismometer		G		P?		G	(M)	
Pressure and Altitude		G M	M	P M				
Urban Heat Islands	(S)	G M	P M	M	P			
Air Convection	(S)	G M	P	P M	P			
The Water Cycle	(S)	G M	P?	(C)	P?			
CO_2 and Photosynthesis		(B)	(B)				B C	
Solar Panels				(M)	(P)	G		P
Physics of the Atmosphere					(P)	G	P M	M
The Beer–Lambert Law						(C)	M	C* P*
Explore tools								
Measurement Notebook		G M	P M	M		G		
Weather Station		G M	P M	P M	M	G	P	
Air Quality		G	M	B	P	G		
Solar Energy			P	M	P	G		P
Seismology		G		P M	M	G		
Heat Islands with Several Boxes		G M	P M	M	P	G		
The Greenhouse Effect		G			P	G C	C	P
Spectroscopy						C	M	C* P*
Games								
Among the Clouds	S M		P					
Earthquake!		G				G		P
Resonance								P (XI)
Gyro Target				B		P		

Subjects: S = primary science (Mathematics and Environmental Exploration in grade II, Natural Sciences in grades III–IV); M = Mathematics; P = Physics; B = Biology; C = Chemistry; G = Geography. **(P)** = as a demonstration or an application only, not a curriculum lesson; **P*** = enrichment or olympiad preparation, not the core curriculum; **P?** = probable link, check the programme in force.

4.3 Specific competences

The plan gives no codes for specific competences (*competențe specifice*). The codes differ from subject to subject and from grade to grade, and a wrong code copied into a teaching plan does more harm than none. Every guide describes the activity's central competence in words, in its "Where this fits the curriculum" section; take the code from the subject's programme:

- **lower secondary** — the school programmes for grades V–VIII approved by order OMEN no. 3393/2017 (for example, the school programme for Physics, grades VI–VIII; Biology, Chemistry, Geography and Mathematics are in the same order);

- **primary** — Natural Sciences, grades III–IV (OMEN no. 5003/2014), and Mathematics and Environmental Exploration, the preparatory grade and grades I–II (OMEN no. 3418/2013);
- **upper secondary** — the programme in force for the grade in question. The upper-secondary programmes are being replaced, so check which one applies to your class.

What all the activities share, in any science programme, is **scientific investigation**: observing, measuring with an instrument, recording and processing data, drawing a conclusion and communicating it. Place them there first, then under the subject content in the table above.

5. How much time does it take?

Three realistic scales of adoption. Choose one and plan equipment accordingly.

Scale	Total time	What it looks like
Taster	1 lesson (50 min)	One game to hook the class (10 min), one Junior exercise done properly with predictions (25 min), discussion (15 min). Requires no preparation beyond charging a box.
Unit	6 lessons over 2–3 weeks	A themed sequence: for example air quality — The Air We Breathe, Indoor Pollution, a week of Air Quality monitoring, then CO_2 and Photosynthesis, ending with pupil presentations.
Full programme	20–30 lessons across a school year	The progression in the next section, one activity every week or fortnight, with two extended projects and a science-fair style showcase.

A single guided experiment occupies 15–50 minutes of app time, but plan **a full 50-minute lesson** around it: predictions, setup, measurement, export and discussion. The durations in the catalogue are the measurement phase only. The exception is **CO_2 and Photosynthesis**: the measurement alone needs at least 45 minutes, so plan a double lesson or seal the jar before the lesson starts.

6. A year-long progression

The sequence below builds skills deliberately. Each stage ends with an open-ended project in which pupils choose their own question — that is where the earlier structured work pays off.

The three stages each correspond to roughly a third of the year and can be laid over the school's own learning modules, however these are divided.

6.1 Stage 1: measuring and comparing

1. **Among the Clouds** (game) — introduce the hardware playfully. Pupils discover that the box knows how it is being held.
2. **Light is Everywhere** — the first real measurement. Skill: reading a value and comparing places.
3. **The Magic Thermometer** — skill: repeated measurement, recognising that values change over time.
4. **The Air We Breathe** — skill: measuring something invisible. First encounter with a sensor that reveals what the senses cannot.
5. **Measurement Notebook** (tool) — skill: systematic labelled recording. Pupils design their own comparison of five places.

6. **Project 1 (2 lessons):** a survey of the school. Groups choose a quantity, measure it in eight locations, and present a ranked chart.

6.2 Stage 2: relationships and variables

1. **Indoor Pollution** — skill: interpreting a trend over time and connecting it to a cause (occupancy, ventilation).
2. **Air Convection** — skill: controlling a variable, measuring at different heights.
3. **Pressure and Altitude** — skill: a systematic series and a near-linear relationship, about 0.12 hPa per metre. Best done in a stairwell.
4. **The Water Cycle** — skill: two quantities changing together. The application computes the dew point, and pupils compare it with the one found with a chilled spoon.
5. **Urban Heat Islands** (the exercise), then **Heat Islands with Several Boxes** (the tool) — skill: field measurement and fair comparison between sites.
6. **Earthquake!** (game) — a change of pace, with a serious debrief on earthquake safety.
7. **Project 2 (3 lessons):** a week-long monitoring campaign. Groups run Air Quality or Weather Station daily, export each day, and compare days. This is where day-to-day file comparison becomes real.

6.3 Stage 3: quantitative science

1. **The Seismometer** and the **Seismology** tool — skill: event detection in a noisy signal.
2. **Solar Panels** — skill: a series of tilt angles, taken at the application's own step, and a cosine relationship.
3. **CO_2 and Photosynthesis** — skill: calculating a rate, and inferring an unmeasured quantity from two measurements. At least 45 minutes of measurement.
4. **Physics of the Atmosphere** — skill: applying a formula to real data, and evaluating how well it fits; the scale height comes from the gradient of the $\ln P$ line, so from grade X.
5. **The Beer–Lambert Law** — skill: transforming data with logarithms to obtain a straight line, then fitting a gradient. From grade X, as enrichment.
6. **Resonance** (game, grade XI) — skill: linking period and frequency, and the idea of a natural frequency, in the oscillations lesson.
7. **Project 3 (3 lessons):** an independent investigation of the pupils' own design, written up as a formal report and presented.

7. Themed routes

If you cannot devote a year, these shorter routes each tell a complete story.

Route	Lessons	Activities in order
Air and health	4	The Air We Breathe → Indoor Pollution → Air Quality (week-long) → CO_2 and Photosynthesis.
Climate and energy	5	Light is Everywhere → Solar Energy → Solar Panels → The Greenhouse Effect → Urban Heat Islands.
Earth in motion	4	Among the Clouds → The Box in Balance → The Seismometer → Earthquake! and Resonance.
Weather and atmosphere	5	The Magic Thermometer → Weather Station → The Water Cycle → Pressure and Altitude → Physics of the Atmosphere.
Light and matter	4	Light is Everywhere → Solar Energy → Spectroscopy → The Beer–Lambert Law.

8. The three-strand assessment rubric

Every guide refers to this rubric and none of them repeats it: it is defined here, once, so that twenty-seven versions of it cannot drift apart.

Three strands are worth tracking over a year, because the activities develop them cumulatively:

- **Measurement practice** — does the pupil take repeated readings, record units and conditions, and recognise when a value is not yet stable?
- **Data interpretation** — can the pupil describe a trend, read a gradient, distinguish a real change from noise, and compare two datasets fairly?
- **Scientific reasoning** — does the pupil predict before measuring, explain a discrepancy, identify assumptions, and propose a test rather than an opinion?

Strand	Emerging	Secure	Confident
Measurement practice	Reads a value off the screen and writes it down. Needs reminding about units and repeats.	Repeats the measurement, records the units and conditions, waits for the value to settle, and changes one variable at a time.	Designs the series unprompted: how many repeats are needed, what is held constant and why, and recognises a compromised reading before using it.
Data interpretation	Says which value is larger and in which direction it changed.	Describes the trend of a graph, links it to the likely cause, and compares two series fairly.	Quantifies: gradient, rate, order of magnitude. Separates noise from signal and says how confident the conclusion is.
Scientific reasoning	Makes a prediction when asked and notices whether it held.	Predicts with a reason, explains the mismatch between prediction and result, and proposes a check.	Names the assumptions behind the method, proposes a test that could disprove their own explanation, and separates the instrument's limits from the theory's.

Beware a clash of words. The three columns above are *levels of attainment* and have nothing to do with the *Junior*, *Explorer* and *Scientist* levels of the activities. A middle-school pupil can

be *confident* on a Junior activity; a sixth-former can be *emerging* on their first open investigation. The rubric describes the pupil, not the activity.

How to use it. Not every lesson. Choose **one strand** per activity — each activity's guide says which one shows up most clearly there — and follow that. Over a year, every pupil will have several observations on each strand, which is more than any test gives.

The best form of portfolio is still the pupil's own folder of exported files with annotated graphs, built up over the year. It shows progression more honestly than any mark.

8.1 From the rubric to marks and qualifiers

The rubric describes; it does not mark. When you do need a mark (*notă*, 1–10) or a qualifier (*calificativ*), the table below is a **suggested conversion**, not a rule: the decision stays yours, and a mark should rest on several observations, not on a single lesson. Each strand is scored as follows: *Emerging* = 1 point, *Secure* = 2 points, *Confident* = 3 points, and a pupil who has not yet reached *Emerging* scores 0 on that strand. When you follow all three strands, the points are added (0–9).

What the pupil showed	Mark (V–XII)	Qualifier (II–IV)
One strand followed		
Confident (3 points)	9–10	Very good (FB)
Secure (2 points)	7–8	Good (B)
Emerging (1 point)	5–6	Sufficient (S)
Not yet *Emerging* (0 points)	below 5	Insufficient (I)
All three strands, points added		
9 points	10	Very good (FB)
8 points	9	Very good (FB)
7 points	8	Good (B)
6 points	7	Good (B)
5 points	6	Sufficient (S)
3–4 points	5	Sufficient (S)
0–2 points	below 5	Insufficient (I)

In primary grades the qualifier is given against the performance descriptors of the programme; the rubric adds concrete observations to them, it does not replace them. A "below 5" or "Insufficient" is not a penalty for a lesson with sensors but a signal that the pupil needs support: repeat the observation at the next activity before entering it in the register.

9. Equipment planning

9.1 How many boxes?

Boxes	Class of 25	What is possible
1	Whole class	Demonstration on a projector; games with pupils taking turns; a single running monitor for a week-long project.
3–4	Groups of 6–8	Rotation through stations. Workable, but the groups are large for the more fiddly setups.
6–8	Groups of 3–4	The comfortable minimum for practical work where everyone handles equipment.

Two activities — Air Quality and Heat Islands with Several Boxes — deliberately support several boxes at once and become considerably richer with three or more.

9.2 Consumables and general equipment

The per-activity guides list exactly what each one needs. Across the whole collection, the recurring items worth having in a box on the shelf are: a desk lamp; sheets of paper and card; a torch; coloured filters or gel sheets; food colouring or ink; transparent containers with lids; a jug for warm tap water; ice; cling film; a cloth or blanket for creating darkness; a tape measure and a ruler; a protractor; clipboards; sticky labels; and two or three robust houseplants.

9.3 Before the first lesson

1. Install the application and connect one box, either by USB cable or over WiFi, and confirm live values appear. The *First Steps* document (`docs/en/first_steps.pdf`) shows the steps and what to do when something does not work. The box has three sockets for the four sensors: before the lesson, with the box switched off, plug in the sensor the activity needs (the table in chapter 2 of the technical manual).
2. In **Settings → Box names**, give each box a name the pupils recognise (for example "Group 1"); the name then appears everywhere, always in the same colour. Lay each box flat: acceleration Z should be about +1 g; for a box that shows −1 g, tick **Motion sensor upside down** in the same place (with firmware 2.3.0 or newer it is kept in the box, which must be connected). For a projector, turn on **Projector mode**, which enlarges the charts.
3. Run one activity yourself, end to end, including the export. Twenty minutes spent here prevents most classroom problems.
4. Check where the data is saved. Sessions are written automatically to the `logs` folder next to the application as CSV files, one column per sensor, and are never deleted on their own. A guided experiment also saves the pupils' journal there and, at **Finish**, an Excel file with the measurements of every step. The **Data & Export** page (Scientist and Teacher levels) exports one box or, with **Export all boxes (Excel)**, the whole lesson; a CSV exported with the application in Romanian uses ";" and the decimal comma, so that it opens correctly in a Romanian Excel. Decide *now* where pupils will keep their files, because day-to-day comparison depends on it.
5. **Print the pupil sheet** for the activity you have chosen — it is a separate document from the guide, two pages long, meant to be printed double-sided, one sheet per child.

Programme-level pitfalls

Treating the sensor as an oracle. Every reading has a warm-up period, a noise level and a failure mode. Teach pupils to ask whether a number is plausible before they trust it. A sensor that disagrees with expectation is sometimes making a discovery and sometimes simply wrong; distinguishing the two is the scientific skill.
Playing the games without debriefing them. Twenty minutes of *Earthquake!* without the discussion afterwards teaches pupils that earthquakes are entertaining, and may leave the door-frame myth intact. **The debrief is not optional.**
Skipping the prediction. It takes 90 seconds and it is the single highest-value habit in the whole programme.
Losing the data. Exports are per session. Establish a file-naming and storage convention in the first lesson, before there is anything to lose.
Over-scripting. The exercises are structured on purpose, but the lasting learning tends to come from the open projects. Protect that time.

General notes

Individual guides carry activity-specific safety information; read the relevant one before teaching, and every pupil sheet has a safety line right under the name. Across the collection, the recurring considerations are trailing cables and power leads; lamps that become hot; warm water (from the tap, poured by the teacher, never from a kettle); electrical equipment near water — **the box is not waterproof**; supervision ratios and route planning for outdoor surveys; and secure handling of the boxes themselves during motion games, where a dropped unit is the most likely damage. Follow your own institution's risk-assessment procedures, which take precedence over anything written here.

10. Where to find things

Item	Location
Teacher guides	`docs/ro/ghiduri_profesor` in Romanian and `docs/en/teacher_guides` in English, one PDF per activity, named after its title (*The Box in Balance* is `box_in_balance.pdf`).
Pupil sheets	`docs/ro/fise_elev` and `docs/en/student_sheets`, with the same name as the guide, two pages each, to be printed double-sided.
Activities by grade and subject	A page to pin up, with the file name of every guide: `docs/en/activities_by_grade_and_subject.pdf`.
First steps	`docs/en/first_steps.pdf` — installing, connecting, common problems.
Curriculum plan	`docs/en/curriculum_plan.pdf` — this document.
Technical manual	`docs/ro/manual_tehnic.pdf` and `docs/en/technical_manual.pdf` — sensors, protocol, file formats.
In the ready-to-run package	All the documents above sit in `MagicBox\docs\ro` and `MagicBox\docs\en`, next to the application.
Automatically saved data	The `logs` folder next to the application, as CSV files named by box and date; high-rate motion data is written to a separate file ending in `_imu`. The pupil journal from an exercise (predictions and observations) is saved there too, as a `_journal.txt` file, and as a *Journal* sheet in the exercise's Excel file.
Exported data	Wherever you choose in the Export dialogue; the suggested name carries the box identifier and the date.
Version	Shown on the application's home screen and in Settings. These documents describe version 5.7.